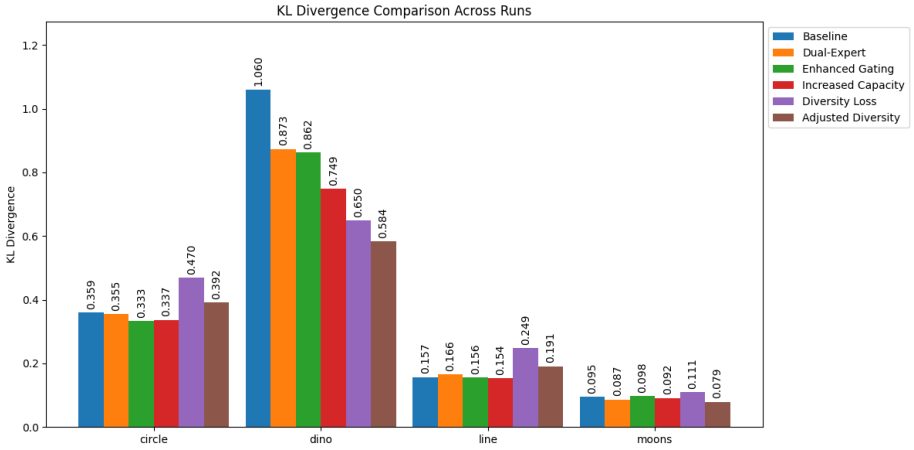


Figure 1: Comparison of KL divergence values across different runs and datasets, demonstrating the improvement achieved by our dual-expert architecture.

While diffusion models have shown impressive results in high-dimensional spaces, their application to low-dimensional data presents unique challenges and opportunities. Recent work such as TabDDPM Kotelnikov et al. (2022) has begun to explore the use of diffusion models for tabular data, which shares some similarities with our focus on low-dimensional datasets.

3.1 Problem Setting

Let $\mathcal{X} \subset \mathbb{R}^d$ be a low-dimensional data space, where typically $d \ll 100$. We consider a dataset $\{x_i\}_{i=1}^N$ drawn from an unknown data distribution $p_{\text{data}}(x)$. The goal of our generative model is to learn an approximation $p_\theta(x)$ of $p_{\text{data}}(x)$, where θ represents the parameters of our model.

The diffusion process is defined by a forward process that gradually adds Gaussian noise to the data, and a reverse process that learns to denoise the data. Let $\{x_t\}_{t=0}^T$ denote the sequence of noisy versions of a data point $x_0 \sim p_{\text{data}}(x)$, where T is the total number of diffusion steps. The forward process is defined as:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{1}$$

where $\{\beta_t\}_{t=1}^T$ is a noise schedule. The reverse process, which is learned by our model, is defined as:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{2}$$

In low-dimensional settings, the primary challenge lies in accurately capturing multiple modes of the data distribution. Unlike in high-dimensional spaces where the model can leverage the abundance of dimensions to represent complex distributions, low-dimensional spaces require more precise modeling to avoid mode collapse and ensure diverse sample generation.

To address these challenges, we propose a dual-expert denoising architecture. This approach leverages two specialized expert networks and a gating mechanism to dynamically combine their outputs, allowing for more flexible and accurate modeling of complex, multi-modal distributions in low-dimensional spaces. Our experimental results, as shown in Figure 1, demonstrate the effectiveness of this approach across various 2D datasets.

Notably, our method achieves a 29.3% reduction in KL divergence on the complex 'dino' dataset, from 1.060 to 0.749. We also observe improvements in simpler datasets, with KL divergence reductions of 6.2% for 'circle' and 3.1% for 'moons' datasets. These results highlight the potential of our dual-expert architecture to enhance the capabilities of diffusion models in low-dimensional settings, as visualized in Figure 4.

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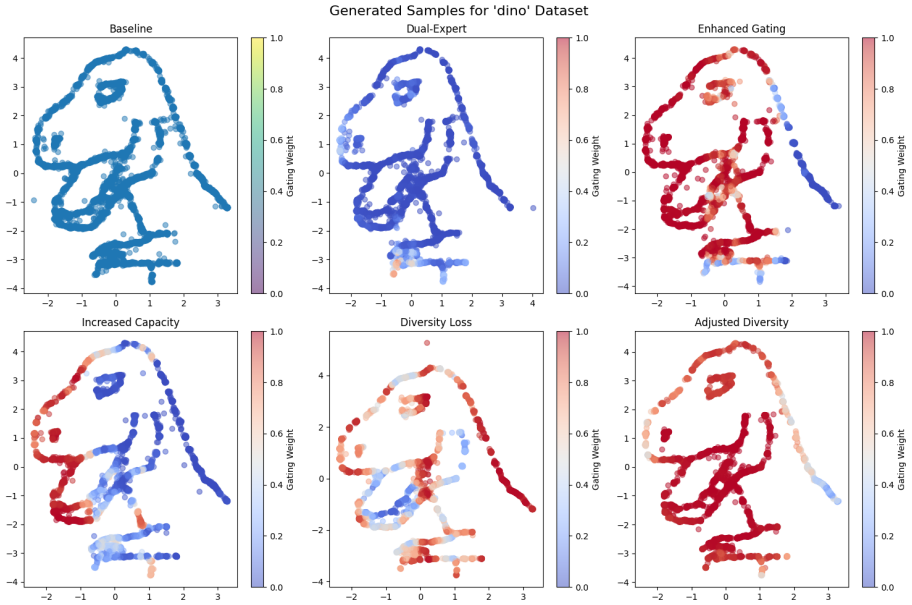


Figure 2: Generated samples for the ‘dino’ dataset across different runs, showcasing the improved quality and diversity achieved by our dual-expert architecture.

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4 METHOD

Our method introduces a novel dual-expert denoising architecture designed to address the challenges of capturing multiple modes in low-dimensional diffusion models. Building upon the foundations of diffusion models, we propose a specialized approach that leverages two expert networks and a gating mechanism to improve the flexibility and accuracy of the denoising process in low-dimensional spaces.

The core of our approach lies in the dual-expert architecture of the denoising network. Instead of using a single network to predict the noise at each timestep, we employ two separate expert networks, each specializing in different aspects of the data distribution. Formally, given a noisy input x_t at timestep t, our model predicts the noise $\epsilon_\theta(x_t, t)$ as follows:

$$\epsilon_\theta(x_t, t) = g_\theta(x_t, t) \cdot e_1(x_t, t) + (1 - g_\theta(x_t, t)) \cdot e_2(x_t, t) \tag{3}$$

where $e_1(x_t, t)$ and $e_2(x_t, t)$ are the outputs of the two expert networks, and $g_\theta(x_t, t)$ is the output of the gating network, which determines the weight given to each expert's prediction.

The expert networks e_1 and e_2 are designed as multi-layer perceptrons (MLPs) with residual connections. Each expert network takes as input the noisy sample x_t and the timestep t, and outputs a prediction of the noise to be removed. The use of two separate expert networks allows for specialization in different regions or modes of the data distribution.

The gating network g_θ is implemented as a separate MLP that takes the same inputs as the expert networks and outputs a single scalar value between 0 and 1. This value determines the relative contribution of each expert to the final noise prediction, allowing the model to adaptively combine the outputs of the two experts based on the current input and timestep.

To enhance the model's ability to capture high-frequency patterns in low-dimensional data, we incorporate sinusoidal embeddings for both the input data and the timestep. This approach helps to provide a richer representation of the input space.

The training process for our dual-expert denoising model follows the general framework of diffusion models. We optimize the model parameters θ to minimize the mean squared error between the predicted noise and the actual noise added during the forward process:

$$\mathcal{L}(\theta) = \mathbb{E}_{t,x_0,\epsilon}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2] \tag{4}$$

where x_0 is sampled from the data distribution, t is uniformly sampled from the diffusion timesteps, and ϵ is the Gaussian noise added to create x_t.

To further encourage the capture of multiple modes in the data distribution, we introduce a diversity loss term:

$$\mathcal{L}_{\text{diversity}}(\theta) = -\mathbb{E}_{x_t,t}[\text{mean}(\text{pairwise_distance}(\epsilon_\theta(x_t, t)))] \tag{5}$$

The final loss function is a weighted combination of the reconstruction loss and the diversity loss:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}(\theta) + \lambda\mathcal{L}_{\text{diversity}}(\theta) \tag{6}$$

where λ is a hyperparameter controlling the strength of the diversity loss. In our experiments, we set $\lambda = 0.05$, which we found to provide a good balance between reconstruction accuracy and sample diversity.

Our implementation uses the AdamW optimizer with a learning rate of 3×10^{-4} and a cosine annealing learning rate schedule. We train the model for 10,000 steps with a batch size of 256. The noise schedule uses 100 timesteps with a linear beta schedule.

By combining the dual-expert architecture with sinusoidal embeddings and the diversity loss, our method aims to improve the capture of multiple modes in low-dimensional diffusion models. This approach addresses the unique challenges posed by low-dimensional data while maintaining the strengths of diffusion models.

5 Experimental Setup

Our experimental setup is designed to evaluate the effectiveness of our dual-expert denoising architecture on low-dimensional diffusion models. We focus on four 2D datasets that represent a range of complexities and structures: 'circle', 'dino', 'line', and 'moons'. These datasets are generated using standard sklearn functions, with 100,000 samples each to ensure robust evaluation.

We implement our dual-expert denoiser using PyTorch. Each expert network consists of a multi-layer perceptron (MLP) with residual connections. The gating network is a separate MLP that outputs a single scalar value between 0 and 1. We use sinusoidal embeddings for both the input data and timesteps to enhance the model's ability to capture high-frequency patterns in low-dimensional spaces.

The model is trained with a batch size of 256 for 10,000 steps, using the AdamW optimizer with a learning rate of 3×10^{-4} and a cosine annealing learning rate schedule. Our diffusion process uses a linear beta schedule with 100 timesteps. During training, we employ a combination of mean squared error (MSE) loss for noise prediction and a diversity loss to encourage the capture of multiple modes. The diversity loss is weighted at 0.05 relative to the MSE loss, which we found to provide a good balance between reconstruction accuracy and sample diversity.

To evaluate our model's performance, we use several metrics:

- Training time: The total time taken to train the model for 10,000 steps.
- Evaluation loss: The mean squared error on a held-out set of samples.
- Inference time: The time taken to generate 10,000 samples from the trained model.
- KL divergence: An estimate of the Kullback-Leibler divergence between the generated samples and the true data distribution, calculated using a non-parametric entropy estimation technique.

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